

Problem

Obtain $g(t)$ if

$$G(s) = \frac{s^3 + 2s + 6}{s(s+1)^2(s+3)}$$

Step-by-step solution

Step 1 of 4

Given function is

$$G(s) = \frac{s^3 + 2s + 6}{s(s+1)^2(s+3)}$$

Using partial fraction expansion to break $G(s)$ down into simple terms

$$\frac{s^3 + 2s + 6}{s(s+1)^2(s+3)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2} + \frac{D}{s+3} \quad (1)$$

Where A, B, C and D are the constants which can be determined by residue method.

$$A = sG(s) \Big|_{s=0}$$

$$A = \frac{s^3 + 2s + 6}{(s+1)^2(s+3)} \Big|_{s=0}$$

$$A = \frac{6}{(1)^2(3)}$$

$$A = 2$$

Step 2 of 4

$$B = \frac{d}{ds} (s+1)^2 G(s) \Big|_{s=-1}$$

$$B = \frac{d}{ds} \left[\frac{s^3 + 2s + 6}{3^2 + 3s} \right] \Big|_{s=-1}$$

$$B = \frac{(s^2 + 3s)[3s^2 + 2] - [s^3 + 2s + 6][2s + 3]}{(s^2 + 3s)^2} \Big|_{s=-1}$$

$$B = \frac{[(-1)^2 + 3(-1)][3(1) + 2] - [(-1)^3 + 2(-1) + 6][2(-1) + 3]}{[(-1)^2 + 3(-1)^2]}$$

$$B = -\frac{13}{4}$$

$$B = -3.25$$

Step 3 of 4

$$C = (s+1)^2 G(s) \Big|_{s=-1}$$

$$C = \frac{s^3 + 2s + 6}{s(s+3)} \Big|_{s=-1}$$

$$C = \frac{(-1)^3 + 2(-1) + 6}{(-1)(-1+3)}$$

$$C = -\frac{3}{2}$$

$$C = -1.5$$

$$D = (s+3) G(s) \Big|_{s=-3}$$

$$D = \frac{s^3 + 2s + 6}{s(s+1)^2} \Big|_{s=-3}$$

$$D = \frac{(-3)^3 + 2(-3) + 6}{(-3)(-3+1)^2}$$

$$D = \frac{-27 - 6 + 6}{(-3)(4)}$$

$$D = \frac{9}{4}$$

$$D = 2.25$$

Step 4 of 4

Thus

$$A = 2, B = 3.25, C = -1.5 \text{ and } D = 2.25$$

Substituting the constant values in equation (1) we get

$$G(s) = \frac{2}{s} - \frac{3.25}{s+1} - \frac{1.5}{(s+1)^2} + \frac{2.25}{s+3}$$

Taking inverse transform on both sides

$$\therefore g(t) = \left[2 - 3.25e^{-t} - 1.5te^{-t} + 2.25e^{-3t} \right] u(t)$$